

QUBO Formulation: Top Eigencentality Node Selection Problem

Top Eigencentality Node Selection Problem

Problem Statement. Given a graph $G = (V, E)$ with $N = |V|$ nodes, a target integer $\tau \in \{1, \dots, N\}$, and pre-computed eigencentality scores c_i for each node $i \in V$, the Top Eigencentality Node Selection Problem seeks to identify a subset of exactly τ nodes that maximizes the sum of their eigencentality scores. Formally, the problem is to find a subset $S \subseteq V$ with cardinality $|S| = \tau$ such that the objective function $\sum_{i \in S} c_i$ is maximized.

Decision Variables. We introduce binary decision variables $x_i \in \{0, 1\}$ for each node $i \in \{0, 1, \dots, N-1\}$. The variable x_i takes the value 1 if node i is selected in the solution, and 0 otherwise.

QUBO Derivation. We convert the maximization problem into a minimization problem suitable for QUBO formulation by negating the objective. The original problem is equivalent to:

$$\min_{x \in \{0,1\}^N} \left(- \sum_{i=0}^{N-1} c_i x_i \right) \quad \text{subject to} \quad \sum_{i=0}^{N-1} x_i = \tau. \quad (1)$$

We incorporate the equality constraint into the objective using a quadratic penalty term with weight $\lambda > 0$. The penalized Hamiltonian $H(x)$ is given by:

$$H(x) = - \sum_{i=0}^{N-1} c_i x_i + \lambda \left(\sum_{i=0}^{N-1} x_i - \tau \right)^2. \quad (2)$$

Expanding the squared penalty term yields:

$$\lambda \left(\sum_{i=0}^{N-1} x_i - \tau \right)^2 = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + \sum_{i \neq j} x_i x_j - 2\tau \sum_{i=0}^{N-1} x_i + \tau^2 \right). \quad (3)$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, we simplify the expression:

$$\lambda \left(\sum_{i=0}^{N-1} x_i + \sum_{i \neq j} x_i x_j - 2\tau \sum_{i=0}^{N-1} x_i + \tau^2 \right) = \lambda \sum_{i=0}^{N-1} (1 - 2\tau) x_i + \lambda \sum_{i \neq j} x_i x_j + \lambda \tau^2. \quad (4)$$

Noting that $\sum_{i \neq j} x_i x_j = 2 \sum_{0 \leq i < j \leq N-1} x_i x_j$, the penalty contribution becomes:

$$\lambda \sum_{i=0}^{N-1} (1 - 2\tau) x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda \tau^2. \quad (5)$$

Combining the objective and penalty terms, the full QUBO Hamiltonian is:

$$H(x) = \sum_{i=0}^{N-1} (\lambda(1 - 2\tau) - c_i) x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda \tau^2. \quad (6)$$

Reading off the coefficients from the general form $H(x) = \sum_i Q_{i,i} x_i + \sum_{i < j} Q_{i,j} x_i x_j + c$, we obtain the QUBO parameters:

$$Q_{i,i} = \lambda(1 - 2\tau) - c_i, \quad \forall i \in \{0, \dots, N-1\}, \quad (7)$$

$$Q_{i,j} = 2\lambda, \quad \forall i, j \in \{0, \dots, N-1\}, \quad i < j, \quad (8)$$

$$c = \lambda \tau^2. \quad (9)$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that the global minimum of $H(x)$ corresponds to a feasible solution. Specifically, λ must exceed the maximum possible gain in the objective function achievable by violating the constraint. Since the objective coefficients are c_i , the maximum gain from selecting an additional node (or deviating from the count τ) is bounded by $\max_i c_i$. Thus, setting $\lambda > \max_{i \in \{0, \dots, N-1\}} c_i$ guarantees that any infeasible solution incurs a penalty strictly greater than any potential improvement in the objective value, forcing the optimizer to satisfy $\sum_i x_i = \tau$.

Correctness Argument. Minimizing $H(x)$ over $x \in \{0, 1\}^N$ recovers the optimal solution to the original problem. For any feasible assignment where $\sum_i x_i = \tau$, the penalty term vanishes, and $H(x)$ reduces to the negative objective function. For any infeasible assignment, the penalty term is strictly positive. By the choice of λ , the energy of any infeasible solution exceeds that of the best feasible solution. Consequently, the global minimum of $H(x)$ must occur at a feasible point that minimizes the negative objective, which is equivalent to maximizing the original objective.

Illustrative Example. Consider an instance with $N = 3$ nodes, target $\tau = 1$, and eigen-centrality scores $c_0 = 5, c_1 = 3, c_2 = 4$. We choose $\lambda = 6$, satisfying $\lambda > \max(c_i) = 5$. Substituting these values into the general formulas:

$$Q_{0,0} = 6(1 - 2) - 5 = -11, \quad (10)$$

$$Q_{1,1} = 6(1 - 2) - 3 = -9, \quad (11)$$

$$Q_{2,2} = 6(1 - 2) - 4 = -10, \quad (12)$$

$$Q_{0,1} = Q_{0,2} = Q_{1,2} = 2(6) = 12, \quad (13)$$

$$c = 6(1)^2 = 6. \quad (14)$$

The resulting Hamiltonian is:

$$H(x) = -11x_0 - 9x_1 - 10x_2 + 12x_0x_1 + 12x_0x_2 + 12x_1x_2 + 6. \quad (15)$$

The feasible solutions are the permutations of $(1, 0, 0)$. Evaluating $H(x)$:

$$H(1, 0, 0) = -11 + 6 = -5, \quad (16)$$

$$H(0, 1, 0) = -9 + 6 = -3, \quad (17)$$

$$H(0, 0, 1) = -10 + 6 = -4. \quad (18)$$

The minimum energy is -5 at $x = (1, 0, 0)$, corresponding to selecting node 0 with the maximum score 5.